

Question number	Answer	Mark
4(a)(i)	B 2 3 The only correct answer is B <i>A is incorrect because glucose and fructose are monosaccharides and lactose, maltose and sucrose are disaccharides</i> <i>C is incorrect because glucose and fructose are monosaccharides and lactose, maltose and sucrose are disaccharides</i> <i>D is incorrect because glucose and fructose are monosaccharides and lactose, maltose and sucrose are disaccharides</i>	(1)

Question number	Answer	Mark
4(a)(ii)	B glycosidic The only correct answer is B. <i>A is incorrect because ester bonds join organic acids and alcohols together</i> <i>C is incorrect because hydrogen bonds do not join two monosaccharides together</i> <i>D is incorrect because phosphodiester bonds join phosphate group to an organic alcohol</i>	(1)

Question number	Answer	Additional guidance	Mark
4(a)(iii)	An answer that includes two of the following points: - monosaccharides are not more or less sweet than disaccharides (1) - a disaccharide is the least sweet and a monosaccharide is the sweetest (1) - sucrose is the sweetest disaccharide and fructose is the sweetest monosaccharide (1)	ACCEPT lactose is the least sweet disaccharide and glucose is the least sweet monosaccharide	(2)

Question number	Answer	Additional guidance	Mark
4(a)(iv)	An answer that includes three of the following points: • {sugars / sugar solutions} are tasted (1) • sugars should be the same concentration (1) • mouth is rinsed out between each sugar (1) • the (relative) sweetness is compared to sucrose (1) OR If students describe Benedict's test, allow the following two marks: • add Benedict's solution and heat (1) • rank sugar by {described colour changes / time taken to reach specific colour} (1)	ACCEPT sampled IGNORE amount ACCEPT the sugars are {compared against each other / given a rating}	(3)

Question number	Answer	Additional guidance	Mark
4(b)	An answer that includes the following points: similarities: • both (polymers) composed of a glucose (1) • both contain 1 - 4 glycosidic {bonds / links} (1) differences: • amylose has {1 - 4 glycosidic bonds (only) / no 1 - 6 glycosidic bonds} and amylopectin has (1 - 4 and) 1 - 6 glycosidic bonds (1)	DO NOT piece together unless in same sentence or two linked adjacent sentences ALLOW composed of glucose if a glycosidic bonds are given ACCEPT amylose is {a chain / helical / linear / unbranched} and amylopectin is branched NB 'amylose has 1 - 4 glycosidic bonds and amylopectin has 1 - 4 and 1 - 6 glycosidic bonds' scores mark points 2 and 3	(3)